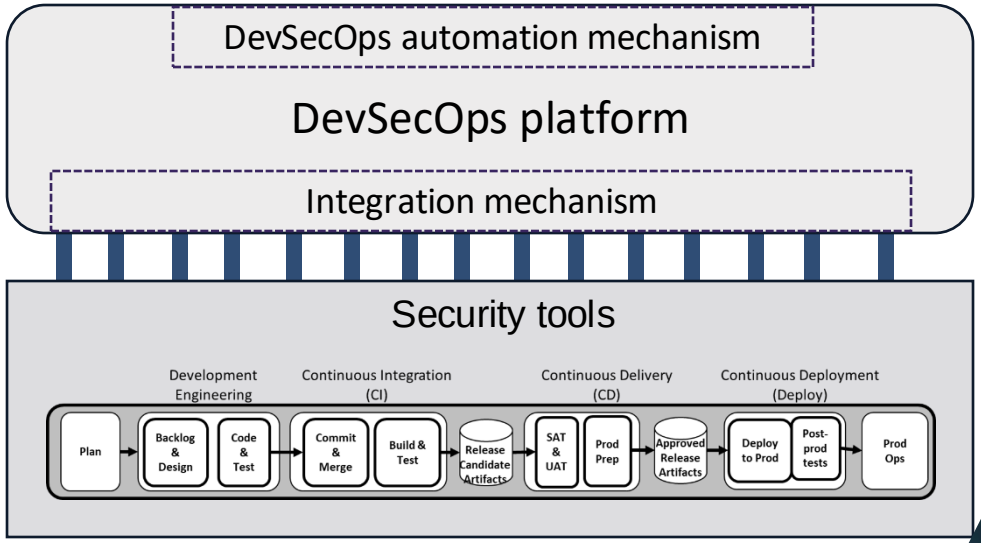
DevSecOps platforms and integrations

DevSecOps platforms

DevSecOps platforms support automation of DevSecOps workflows by integrating security tools for each pipeline into a common ecosystem. This enables seamless, automated security across development, delivery and deployment.

DevSecOps platform choices:

- Do-it-Yourself Platforms (DIY)
- CI/CD Platforms
- Value-Stream Management Platform (VSMP)
- Automated Release Orchestration (ARO) platforms
- platform engineering.



Source: https://mybook.to/HtAAS8



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Do-it-Yourself Platforms (DIY)

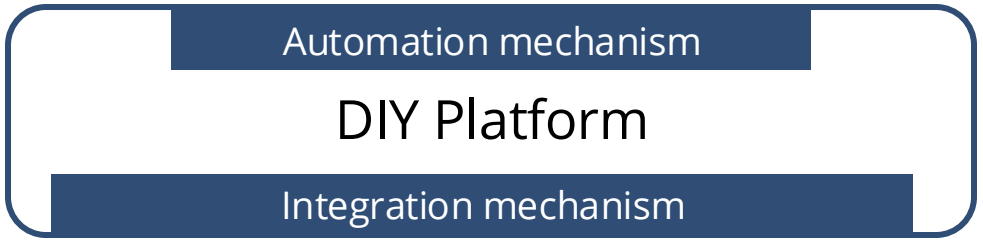
Custom-built platforms using open-source tools or custom development to integrate and automate security processes.

Automation mechanisms:

- Scripting (for example, Python, Bash).
- Generative AI for building and refining custom workflows.
- IaC for automating security configurations

Integration mechanisms:

- APIs for tool integration.
- Custom scripting or coding interfaces to connect ICS tools.



Example products:

- Open-source tools like Terraform, Ansible, and Jenkins.
- Custom-built solutions leveraging tools like Docker and Kubernetes.
- Custom integrations with other platform tools.

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CI/CD Platforms

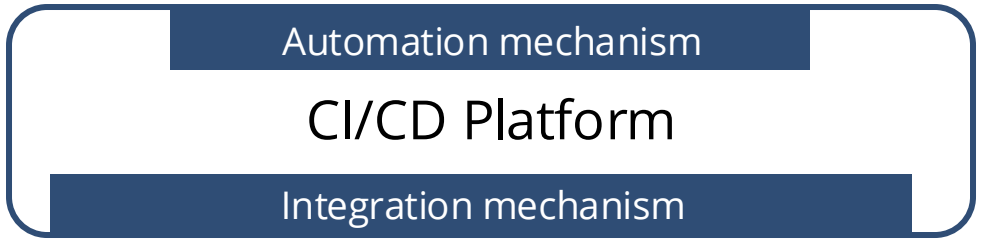
Continuous Integration/Continuous Delivery platforms automate the build, test, and deployment processes.

Automation mechanisms:

- DSL (Domain-Specific Language) for pipeline scripting
- Generative AI for creating secure pipelines
- templates for reusable security workflows.

Integration mechanisms:

- Native plug-ins for ICS tools (for example, vulnerability scanners, security tests).
- APIs to connect external tools.



Example products:

- Jenkins
- CloudBees
- GitLab CI/CD
- Harness
- Azure DevOps
- CircleCI.



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Value Stream Management Platform (VSMP)

Tools that provide end-to-end visibility, management, and optimization of value streams, including security workflows.

Automation mechanisms:

- Generative AI to analyse and optimize value streams for security
- dashboards and workflows to automate security monitoring and reporting.

Integration mechanisms:

- APIs for integrating ICS tools
- data model integration for tracking security metrics.

Automation mechanism

Value Stream Management Platform

Integration mechanism

Example products:

- Tasktop Viz
- Plutora
- Jira Align.



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Automated Release Orchestration (ARO) Platforms

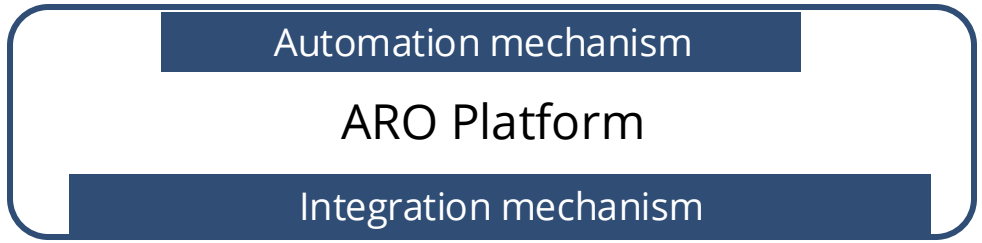
Platforms that automate and coordinate the release of software, ensuring secure and efficient deployments.

Automation mechanisms:

- DSL for scripting release workflows
- automated policy validation and compliance checks during deployment.

Integration mechanisms:

- APIs and plug-ins for integrating DevSecOps tools into release pipelines
- coding interfaces to define security policies in releases.



Example products:

- CloudBees CD/RO
- Microsoft Release Manager
- Octopus Deploy.



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Platform engineering

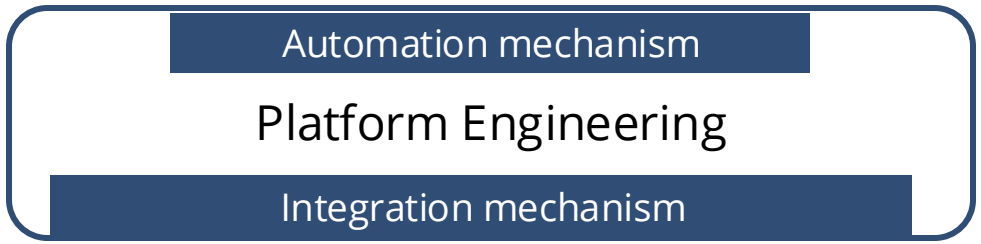
Builds internal platform portals for self-service infrastructure and developer workflows, including integration of DevSecOps tools.

Automation mechanisms:

- scripting and automation via IaC
- Generative AI for building self-service interfaces for ICS tools.

Integration mechanisms:

- templates and APIs for standardizing DevSecOps tool integration
- custom data models to map security workflows.



Example products:

- Backstage (Spotify)
- Humanitec
- HashiCorp Terraform Enterprise.



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DevSecOps platform engineers

DevSecOps pipeline specialists are considered platform engineers:

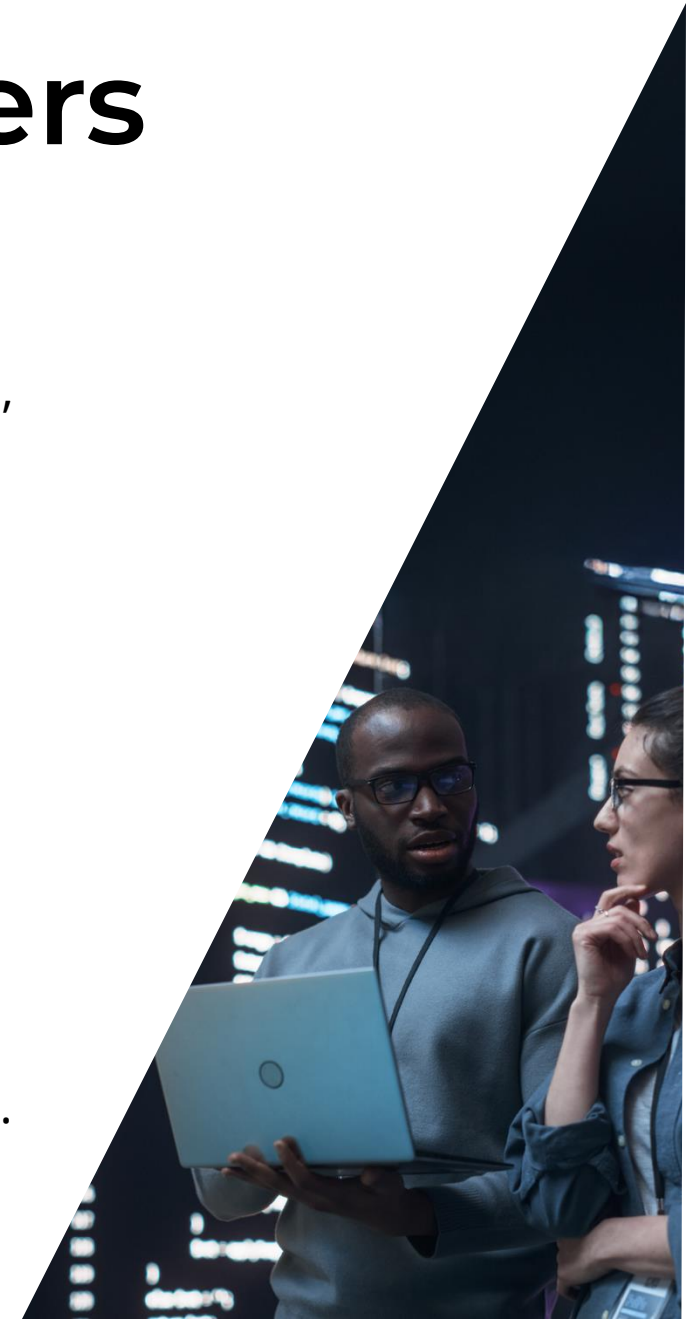
- Knowledge of DevSecOps platforms and tools, strong communicator, keen analytics, decompose complex processes, proficient at optimizing pipelines, team builder.

Duties:

- Develop CI/CD principles.
- Review/modify CI/CD principles, iteratively.
- Maintain CI/CD tools/platforms.
- Automate processes.

Technical skills:

- Write scripts.
- Interpret/write source code.
- Manage infrastructure.
- Be familiar with software packing and version control tools.
- Be familiar with security, vulnerability, monitoring, and code coverage tools.

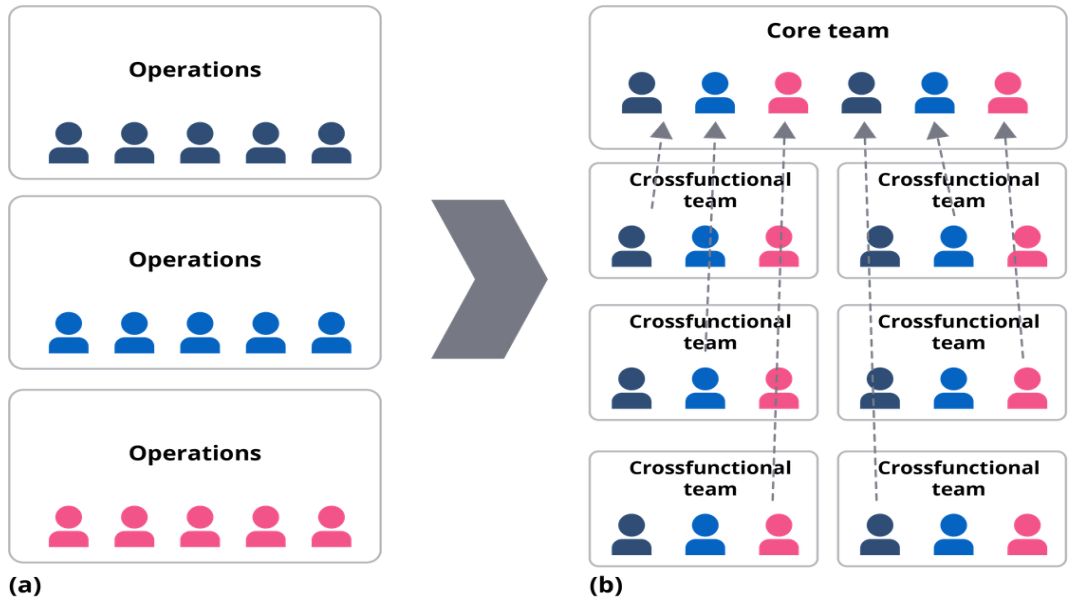


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Integrating people

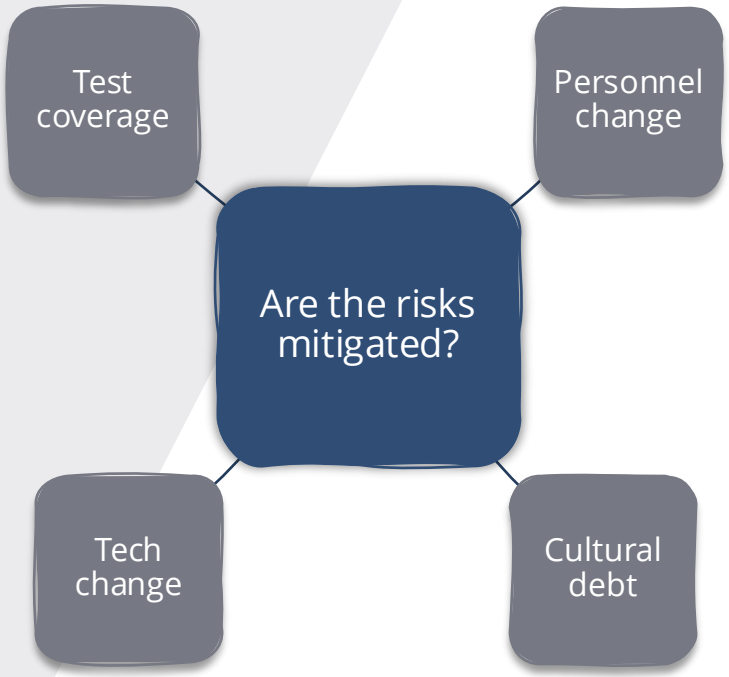
Pipelines bring teams together with shared artifacts.

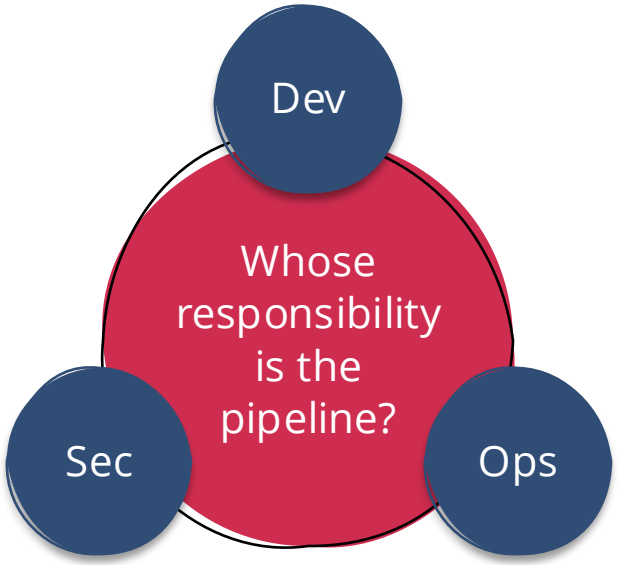
- **Challenge:** choosing the wrong team.
  - Start with cross-functional and project minded.
- **Challenge:** lacking a plan.
  - Roadmaps do not have to be complete, just present.
- **Challenge:** rigid implementation.
  - Guidelines are guidelines and retros create change.
- **Challenge:** not giving importance to the database.
  - Planning with architects and structure for data transition.

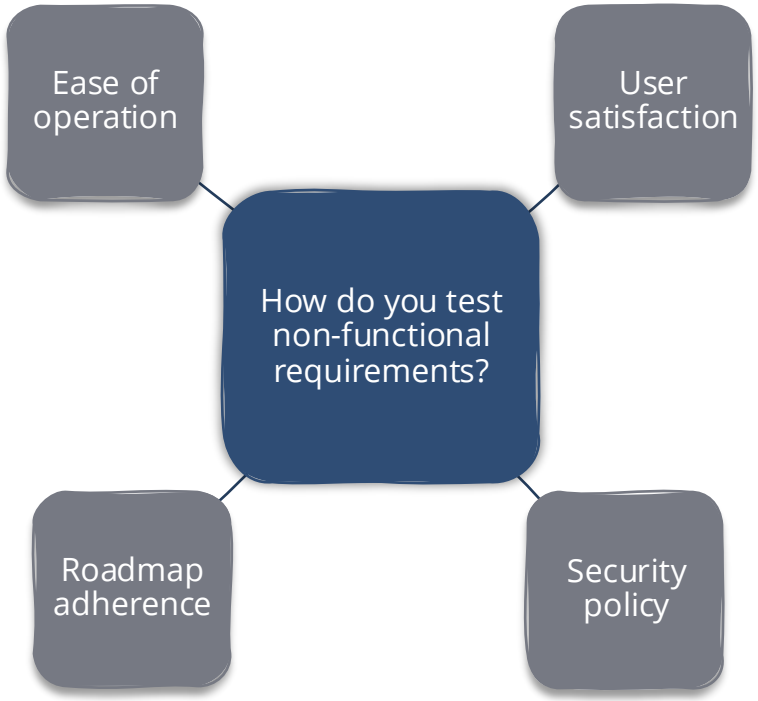


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DevSecOps pipeline responsibilities







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